DEBT SUSTAINABILITY ANALYSIS

A. Debt Sustainability Analysis Without Remittances

External Debt

6. Under the baseline assumptions, all external debt and debt-service ratios remain well below the policy-dependent thresholds throughout the projection period (Figure 1). The present value (PV) of external debt, calculated using a 5 percent discount rate, is expected to climb slightly from 17 percent of GDP in 2013 to 19 percent in 2033 (Table 1a). As production from existing gold mines declines starting in 2015 and growing other exports only partly compensate for that decline, the PV of the external debt-to-exports ratio is projected to increase from 66 percent in 2013 to 128 percent in 2033, below the threshold of 150 percent (Figure 1c, Table 1a). With a 5 percent increase in tax revenue to GDP during the projection period, the PV of the external debt-to-revenue ratio is expected to decrease from 95 percent in 2013 to 84 percent in 2033, remaining, significantly below the threshold of 250 percent (Figure 1d, Table 1a).

7. Mali's external debt sustainability is sensitive to a hardening of financial terms, a reduction in transfers and FDI, and an export shock, limiting the scope for non-concessional borrowing. Under a hardening of financial terms, the PV of debt-to-exports ratio would breach the threshold in 2023 and continue to increase until the end of the projection period to reach about 213 percent in 2033 (Figure 1c, Table 1b, Scenario A2). Under a bound test that reduces FDI and official and private transfers in 2014–15 with the effect of reducing the combined inflows permanently by 10 percent, the PV of the debt-to-exports ratio would exceed the threshold from 2020 until the end of the projection period (Table 1b, Scenario B4). Under a bound test that reduces exports' *growth* temporarily in 2014–15 with the effect of reducing exports *levels* permanently by 10 percent, the PV of the debt-to-exports ratio would reach the threshold at the end of the projection period (Table 1b, Scenario B2).

Public Debt

8. The inclusion of domestic debt does not alter the assessment of Mali's debt sustainability. Given the small size of Mali's domestic debt and the absence of recourse to domestic borrowing in the baseline scenario, the public debt sustainability analysis closely mirrors the external debt sustainability analysis (Figure 2 and Table 2a). The PV of debt-to-GDP ratio stays around 20 percent of GDP during the whole projection period.

B. Debt Sustainability Analysis Including Remittances

9. Mali's external debt sustainability assessment remains similar when remittances are accounted for. Workers' remittances represent a reliable source of foreign exchange in Mali: they averaged 5.2 percent of GDP during the past decade and 6.6 percent of GDP during the past three years. Under the baseline assumptions, all external debt and debt-service ratios remain below the policy-dependent thresholds throughout the projection period (Figure 3, Table 3a). Over 2013–33, the PV of debt-to-GDP plus remittances slightly increases from 16 percent to 18 percent, and the PV of debt-to-exports

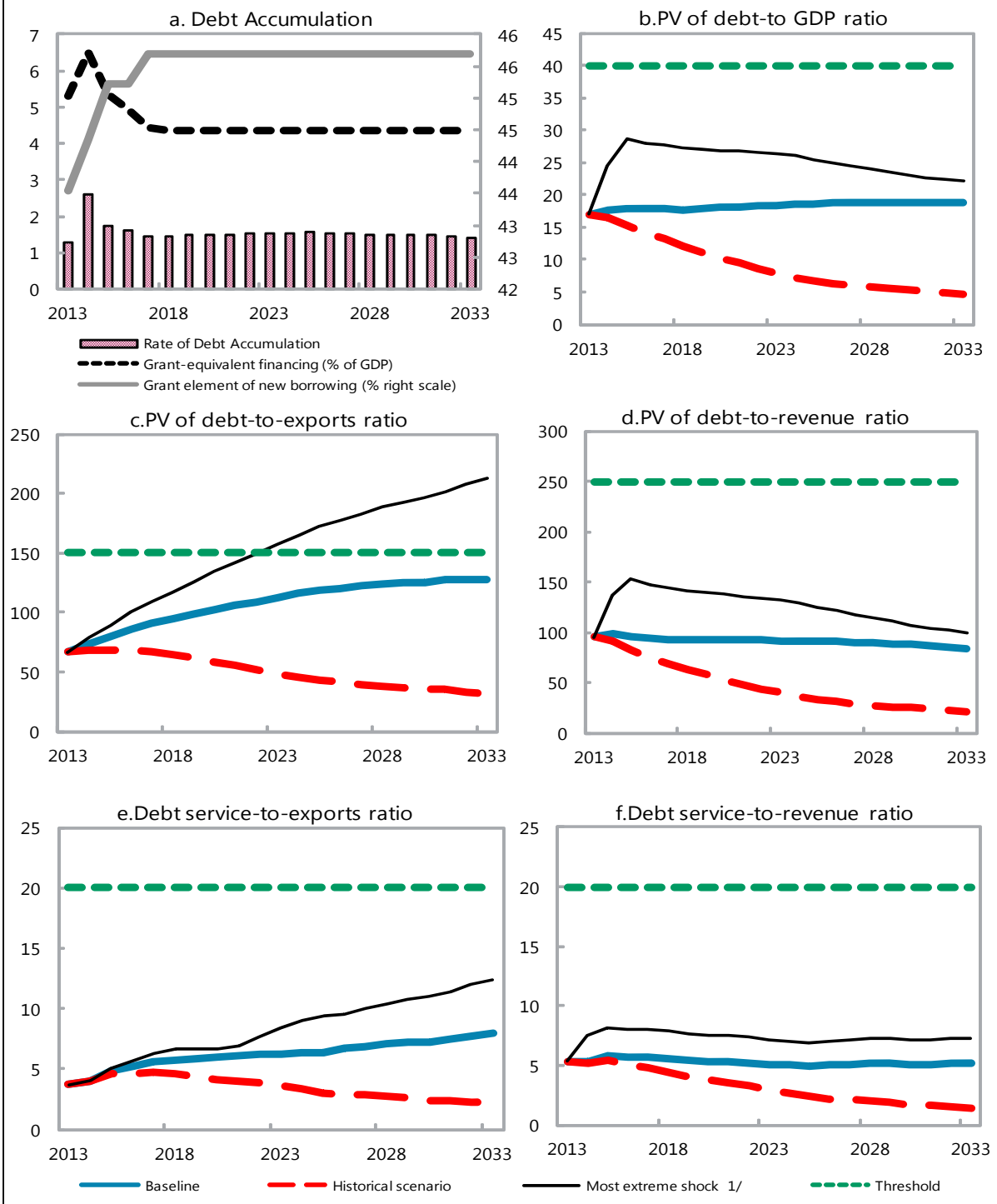
plus remittances increases from 53 percent to 92 percent (Table 3a). Under a hardening of financial terms, the PV of debt-to-exports plus remittances ratio would breach the threshold in 2024, and remain above it until the end of the projection period (Figure 3c, Table 3a, Scenario A2).

10. The Malian authorities agreed with the conclusions of the DSA. They reaffirmed their commitment to continue meeting their financing need with grants and non concessional loans to the extent possible. However confronted with high needs for investment in infrastructure, they expressed the willingness to finance investment projects carrying high return and/or positive spillovers to the rest of the economy with market financing if concessional finance is not available and provided that the implementation of such projects would remain compatible with debt sustainability.

DEBT DISTRESS CLASSIFICATION AND CONCLUSIONS

11. The DSA indicates that Mali remains at moderate risk of debt distress based on the external debt burden indicators. As in last year's DSA, none of the debt burden thresholds are breached over the 20–year projection period under the baseline scenario, and debt sustainability remains mostly sensitive to a hardening of financing terms, a reduction in transfers and FDI, and an export shock owing to the export concentration on gold,. However, given the expected decline in gold exports in the medium term, and the uncertain prospects for export diversification, Mali's debt sustainability needs to remain under close scrutiny. The above mentioned factors necessitate recommending that the government continue to meet its external financing needs mostly with grants and concessional loans. Increasing other exports to compensate the expected decline in gold exports will be critical to maintain external debt sustainability.

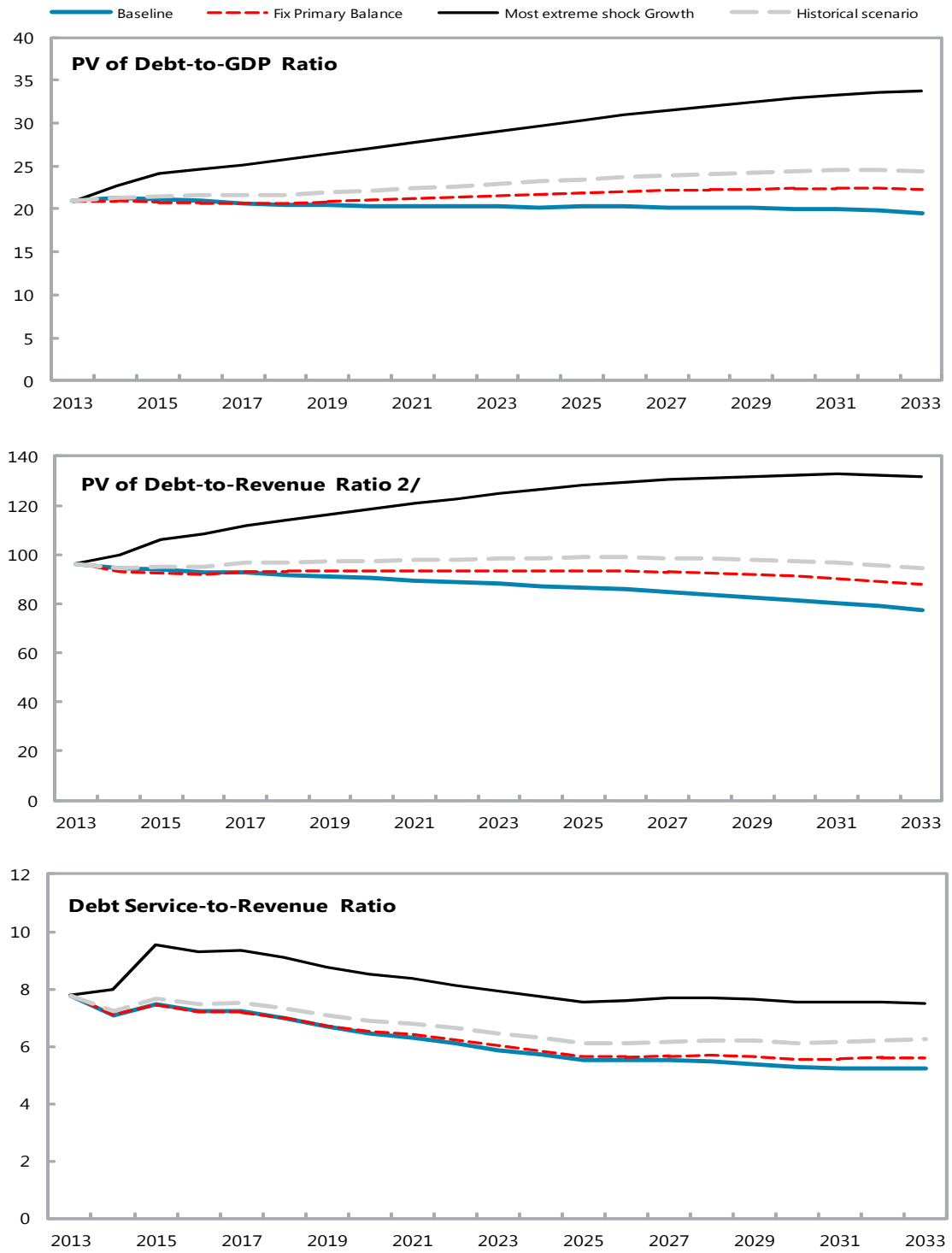
Figure 1. Mali: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2024. In figure b. it corresponds to a Non-debt flows shock; in c. to a Terms shock; in d. to a Non-debt flows shock; in e. to a Terms shock and in figure f. to a One-time depreciation shock

Figure 2.Mali: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1a.Mali: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**24.3**	**24.4**	**26.0**			**26.1**	**27.4**	**27.8**	**27.9**	**28.0**	**28.0**		**29.2**	**30.1**	
of which: public and publicly guaranteed (PPG)	24.3	24.4	26.0			26.1	27.4	27.8	27.9	28.0	28.0		29.2	30.1	
Change in external debt	1.7	0.1	1.7			0.0	1.4	0.3	0.2	0.0	0.1		0.2	-0.1	
Identified net debt-creating flows	5.6	-4.6	1.1			0.9	1.9	1.5	1.3	1.4	1.4		1.6	1.3	
Non-interest current account deficit	**10.6**	**3.0**	**3.1**	**4.5**	**3.2**	**5.5**	**7.3**	**6.8**	**6.3**	**6.2**	**6.0**		**5.4**	**3.7**	4.8
Deficit in balance of goods and services	17.0	9.8	6.5			26.5	21.8	16.6	15.6	12.8	12.8		13.1	12.7	
Exports	26.0	26.3	31.1			25.6	23.9	22.2	20.7	19.6	18.7		16.3	14.6	
Imports	43.0	36.1	37.6			52.1	45.7	38.8	36.2	32.4	31.5		29.4	27.3	
Net current transfers (negative = inflow)	-8.8	-8.0	-7.7	-7.1	1.4	-23.6	-15.0	-10.4	-9.7	-7.0	-7.1		-7.3	-7.6	-7.4
of which: official	-2.1	-1.6	-0.9			-17.2	-9.0	-4.7	-4.3	-1.9	-1.9		-1.9	-1.9	
Other current account flows (negative = net inflow)	2.5	1.2	4.3			2.5	0.4	0.6	0.4	0.4	0.3		-0.5	-1.4	
Net FDI (negative = inflow)	**-4.2**	**-5.2**	**-3.0**	**-3.4**	**2.1**	**-3.6**	**-4.1**	**-4.2**	**-3.9**	**-3.9**	**-3.7**		**-2.7**	**-1.3**	-2.3
Endogenous debt dynamics 2/	**-0.8**	**-2.5**	**1.0**			**-0.9**	**-1.3**	**-1.1**	**-1.1**	**-0.9**	**-0.9**		**-1.0**	**-1.1**	
Contribution from nominal interest rate	0.3	0.4	0.4			0.3	0.3	0.3	0.3	0.3	0.3		0.4	0.4	
Contribution from real GDP growth	-1.3	-0.6	0.1			-1.2	-1.6	-1.4	-1.4	-1.3	-1.3		-1.4	-1.5	
Contribution from price and exchange rate changes	0.2	-2.2	0.6			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.9**	**4.7**	**0.6**			**-0.8**	**-0.5**	**-1.1**	**-1.2**	**-1.4**	**-1.3**		**-1.4**	**-1.5**	
of which: exceptional financing	-0.2	-0.3	-0.1			-0.5	-0.4	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	17.1			17.0	17.7	17.8	17.8	17.8	17.7		18.4	18.7	
In percent of exports	...	...	54.9			66.3	73.9	79.9	86.1	90.7	94.9		112.6	127.6	
PV of PPG external debt	**...**	**...**	**17.1**			**17.0**	**17.7**	**17.8**	**17.8**	**17.8**	**17.7**		**18.4**	**18.7**	
In percent of exports	**...**	**...**	**54.9**			**66.3**	**73.9**	**79.9**	**86.1**	**90.7**	**94.9**		**112.6**	**127.6**	
In percent of government revenues	**...**	**...**	**99.3**			**95.4**	**97.8**	**95.1**	**93.3**	**92.9**	**92.2**		**91.8**	**83.6**	
Debt service-to-exports ratio (in percent)	**3.7**	**4.9**	**3.5**			**3.7**	**4.0**	**4.9**	**5.2**	**5.6**	**5.7**		**6.3**	**7.9**	
PPG debt service-to-exports ratio (in percent)	**3.7**	**4.9**	**3.5**			**3.7**	**4.0**	**4.9**	**5.2**	**5.6**	**5.7**		**6.3**	**7.9**	
PPG debt service-to-revenue ratio (in percent)	**5.6**	**7.4**	**6.3**			**5.3**	**5.3**	**5.8**	**5.7**	**5.7**	**5.6**		**5.1**	**5.2**	
Total gross financing need (Billions of U.S. dollars)	0.7	-0.1	0.1			0.3	0.5	0.5	0.5	0.6	0.6		0.9	1.9	
Non-interest current account deficit that stabilizes debt ratio	8.9	3.0	1.4			5.4	5.9	6.4	6.1	6.2	5.9		5.2	3.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.8	2.7	-0.4	4.3	2.3	5.1	6.6	5.5	5.5	4.9	4.9	5.4	5.1	5.2	5.1
GDP deflator in US dollar terms (change in percent)	-0.7	10.2	-2.3	8.2	8.7	4.3	4.1	3.5	3.3	3.2	3.2	3.6	2.4	1.8	2.3
Effective interest rate (percent) 5/	1.2	1.7	1.4	1.5	0.3	1.2	1.3	1.3	1.3	1.3	1.3	1.3	1.4	1.4	1.4
Growth of exports of G&S (US dollar terms, in percent)	15.0	14.3	15.4	12.7	15.2	-9.8	3.8	1.6	1.3	2.5	3.4	0.5	5.1	6.3	5.8
Growth of imports of G&S (US dollar terms, in percent)	30.9	-5.0	1.3	15.2	22.9	52.0	-2.6	-7.2	1.8	-3.2	5.1	7.6	5.7	6.6	6.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	43.5	44.3	45.2	45.2	45.7	45.7	45.0	45.7	45.7	45.7
Government revenues (excluding grants, in percent of GDP)	17.3	17.2	17.2			17.8	18.1	18.7	19.1	19.1	19.2		20.0	22.3	20.7
Aid flows (in Billions of US dollars) 7/	0.6	0.8	0.1			0.7	1.0	0.9	0.9	0.9	0.9		1.3	2.7	
of which: Grants	0.3	0.4	0.0			0.5	0.6	0.5	0.5	0.5	0.5		0.8	1.6	
of which: Concessional loans	0.3	0.3	0.0			0.2	0.4	0.3	0.3	0.4	0.4		0.5	1.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.3	6.5	5.3	4.9	4.4	4.3		4.3	4.3	4.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			75.1	72.0	74.1	73.7	73.6	73.2		73.2	73.2	73.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	9.4	10.7	10.4			11.4	12.7	13.8	15.1	16.3	17.7		25.4	52.5	
Nominal dollar GDP growth	5.0	13.2	-2.7			9.7	11.0	9.2	9.0	8.2	8.3	9.2	7.7	7.1	7.5
PV of PPG external debt (in Billions of US dollars)			1.8			2.0	2.3	2.5	2.7	2.9	3.1		4.7	10.1	
(PVt-PVt-1)/GDPt-1 (in percent)						1.3	2.6	1.7	1.6	1.4	1.5	1.7	1.5	1.4	1.5
Gross workers' remittances (Billions of US dollars)	0.6	0.7	0.7			0.7	0.8	0.8	0.8	0.8	0.9		1.4	3.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	16.0			16.0	16.7	16.8	16.9	16.9	16.9		17.4	17.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	45.1			53.2	59.2	63.7	68.3	71.8	74.1		84.4	91.6	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.9			3.0	3.2	3.9	4.2	4.4	4.5		4.7	5.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033
(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	17	18	18	18	18	18	**18**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	17	16	15	14	13	12	**8**	5
A2. New public sector loans on less favorable terms in 2013-2033 /2	17	19	20	21	21	22	**26**	31
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	17	19	19	19	19	19	**20**	21
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	17	19	20	20	20	20	**20**	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	17	19	19	19	19	19	**20**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	17	25	29	28	28	27	**26**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	17	24	28	27	27	27	**26**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	17	25	25	25	25	25	**26**	27
PV of debt-to-exports ratio								
Baseline	66	74	80	86	91	95	**113**	128
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	66	69	69	69	67	65	**49**	32
A2. New public sector loans on less favorable terms in 2013-2033 /2	66	79	89	100	109	117	**158**	213
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	66	74	80	86	91	95	**113**	131
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	66	83	99	106	112	116	**136**	150
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	66	74	80	86	91	95	**113**	131
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	66	103	128	136	141	146	**162**	152
B5. Combination of B1-B4 using one-half standard deviation shocks	66	97	113	120	125	129	**145**	141
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	66	74	80	86	91	95	**113**	131
PV of debt-to-revenue ratio								
Baseline	95.4	98	95	93	93	92	**92**	84
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	95	91	82	75	69	63	**40**	21
A2. New public sector loans on less favorable terms in 2013-2033 /2	95	105	106	108	111	114	**128**	139
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	95	103	103	101	101	100	**100**	93
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	95	103	106	104	103	102	**100**	88
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	95	103	104	102	101	101	**100**	93
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	95	136	153	147	144	141	**132**	99
B5. Combination of B1-B4 using one-half standard deviation shocks	95	135	149	144	141	139	**131**	102
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	95	138	134	132	131	130	**130**	120

Table 1b.Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (Continued)
(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
Debt service-to-exports ratio								
Baseline	4	4	5	5	6	6	**6**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	4	5	5	5	5	**4**	2
A2. New public sector loans on less favorable terms in 2013-2033 /2	4	4	5	6	6	7	**8**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	4	5	5	6	6	**6**	8
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	4	6	6	6	7	**7**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	4	5	5	6	6	**6**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	4	6	6	7	7	**7**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	6	6	6	**7**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	4	5	5	6	6	**6**	8
Debt service-to-revenue ratio								
Baseline	5	5	6	6	6	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	5	5	5	5	5	4	**3**	1
A2. New public sector loans on less favorable terms in 2013-2033 /2	5	5	6	6	6	7	**7**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	5	6	6	6	6	6	**6**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	5	6	6	6	6	**5**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	5	6	6	6	6	6	**6**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	5	5	7	7	7	7	**6**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	6	7	7	7	7	**6**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	5	7	8	8	8	8	**7**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	**44**	44

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a.Mali: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	28.7	29.1	30.1			30.0	31.1	31.1	31.0	30.9	30.7		31.1	31.0	
of which: foreign-currency denominated	24.3	24.4	26.0			26.1	27.4	27.8	27.9	28.0	28.0		29.2	30.1	
Change in public sector debt	4.0	0.4	1.0			-0.1	1.0	0.1	-0.1	-0.2	-0.1		0.1	-0.2	
Identified debt-creating flows	1.8	0.6	-0.3			-0.9	0.2	0.1	-0.1	-0.1	-0.1		0.0	-0.1	
Primary deficit	2.3	3.1	0.7	2.2	1.0	2.2	2.9	2.2	2.0	1.8	1.8	2.1	1.8	1.8	1.8
Revenue and grants	20.1	21.1	17.5			21.7	22.5	22.5	22.5	22.2	22.2		23.0	25.3	
of which: grants	2.9	3.8	0.2			3.9	4.5	3.8	3.5	3.1	3.0		3.0	3.0	
Primary (noninterest) expenditure	22.4	24.1	18.1			23.9	25.4	24.6	24.5	24.0	24.0		24.8	27.1	
Automatic debt dynamics	0.3	-1.4	-0.9			-1.7	-2.3	-2.1	-2.1	-1.9	-1.9		-1.7	-1.9	
Contribution from interest rate/growth differential	-1.2	-0.6	0.2			-1.4	-1.9	-1.7	-2.4	-1.6	-1.5		-1.6	-1.3	
of which: contribution from average real interest rate	0.1	0.2	0.0			0.1	0.0	-0.1	-0.8	-0.1	-0.1		-0.1	0.3	
of which: contribution from real GDP growth	-1.4	-0.8	0.1			-1.5	-1.9	-1.6	-1.6	-1.5	-1.4		-1.5	-1.5	
Contribution from real exchange rate depreciation	1.5	-0.8	-1.0			-0.3	-0.5	-0.4	0.3	-0.3	-0.3		...	...	
Other identified debt-creating flows	-0.7	-1.1	-0.1			-1.4	-0.4	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			-1.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.7	-1.1	-0.1			-0.5	-0.4	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.2	-0.2	1.3			0.8	0.9	0.0	0.0	-0.1	-0.1		0.0	-0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	21.2			20.9	21.3	21.1	20.9	20.7	20.4		20.2	19.6	
of which: foreign-currency denominated	...	...	17.1			17.0	17.7	17.8	17.8	17.8	17.7		18.4	18.7	
of which: external	...	...	17.1			17.0	17.7	17.8	17.8	17.8	17.7		18.4	18.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.4	6.4	4.8			6.4	6.8	6.0	5.6	5.2	5.1		4.3	3.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	121.2			96.3	94.5	94.0	92.7	93.0	91.9		88.0	77.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	122.9			117.6	117.9	113.1	109.7	108.0	106.3		101.2	87.6	
of which: external 3/	...	...	99.3			95.4	97.8	95.1	93.3	92.9	92.2		91.8	83.6	
Debt service-to-revenue and grants ratio (in percent) 4/	7.6	10.8	10.9			7.8	7.1	7.5	7.2	7.2	7.0		5.9	5.2	
Debt service-to-revenue ratio (in percent) 4/	8.9	13.2	11.0			9.5	8.8	9.0	8.6	8.4	8.0		6.8	5.9	
Primary deficit that stabilizes the debt-to-GDP ratio	-1.7	2.6	-0.3			2.3	1.8	2.1	2.1	1.9	1.9		1.7	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.8	2.7	-0.4	4.3	2.3	5.1	6.6	5.5	5.5	4.9	4.9	5.4	5.1	5.2	5.1
Average nominal interest rate on forex debt (in percent)	1.2	1.7	1.4	1.5	0.3	1.2	1.3	1.3	1.3	1.3	1.3	1.3	1.4	1.4	1.4
Average real interest rate on domestic debt (in percent)	2.5	1.5	0.4	0.2	2.6	4.9	2.3	2.5	2.4	2.4	2.4	2.8	2.2	1.8	2.2
Real exchange rate depreciation (in percent, + indicates depreciation)	6.9	-3.4	-4.2	-4.5	9.1	-1.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.2	5.1	5.7	4.6	1.8	1.2	2.3	2.1	2.2	2.2	2.2	2.0	2.4	2.8	2.4
Growth of real primary spending (deflated by GDP deflator, in percent)	-7.3	10.7	-25.1	2.2	14.3	38.5	13.3	2.4	5.0	2.7	5.0	11.2	5.8	6.3	6.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	43.5	44.3	45.2	45.2	45.7	45.7	45.0	45.7	45.7	45.7

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Mali: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

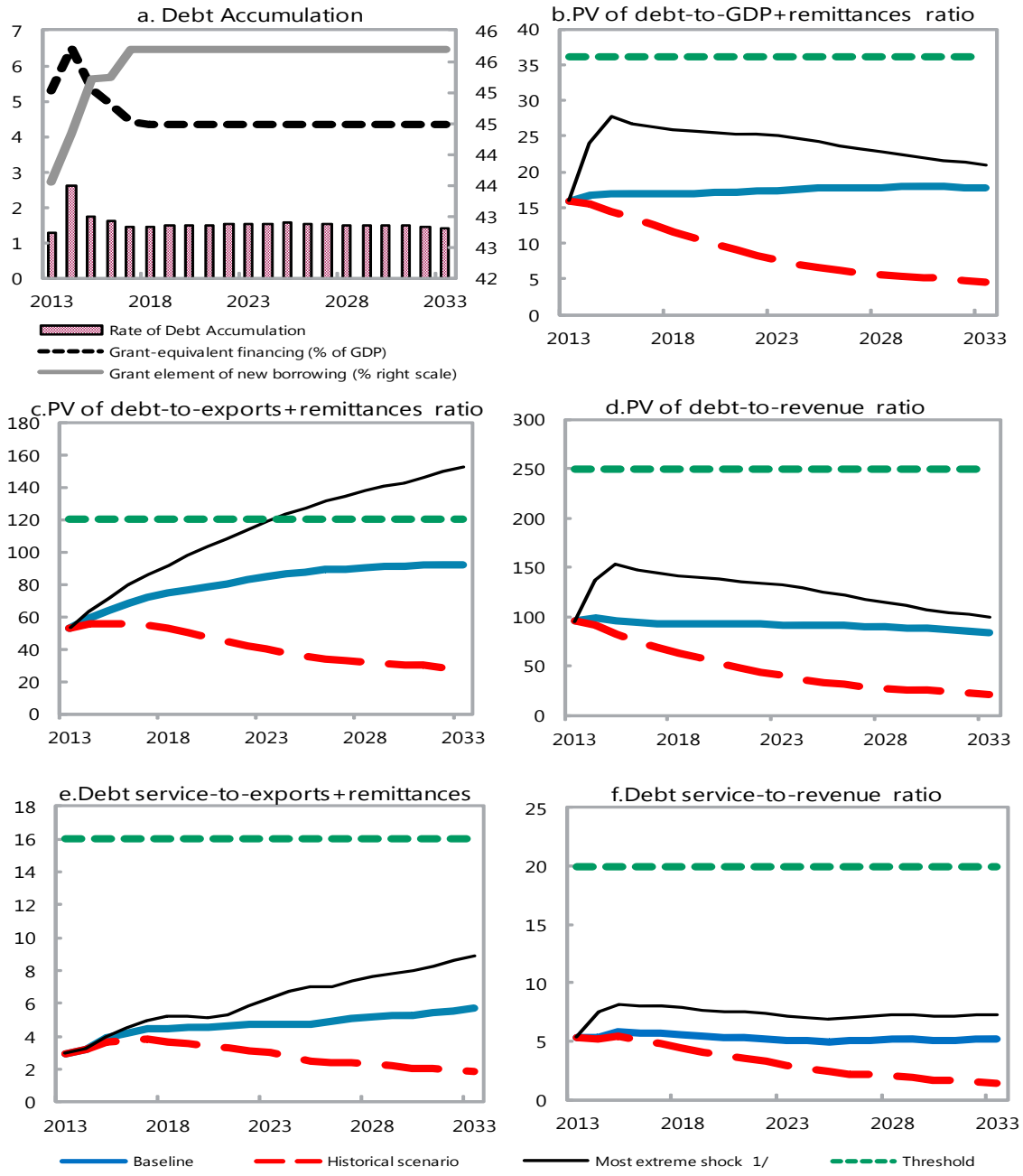
	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	21	21	21	21	21	20	20	20
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	21	21	21	22	22	22	23	24
A2. Primary balance is unchanged from 2013	21	21	21	21	21	21	22	22
A3. Permanently lower GDP growth 1/	21	21	21	21	22	22	24	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	21	23	24	25	25	26	29	34
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	21	21	22	22	21	21	21	20
B3. Combination of B1-B2 using one half standard deviation shocks	21	22	22	23	23	23	26	29
B4. One-time 30 percent real depreciation in 2014	21	27	26	25	24	23	21	17
B5. 10 percent of GDP increase in other debt-creating flows in 2014	21	27	26	26	25	25	24	22
PV of Debt-to-Revenue Ratio 2/								
Baseline	96	94	94	93	93	92	88	77
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	96	94	95	95	97	97	98	94
A2. Primary balance is unchanged from 2013	96	93	92	92	93	93	93	88
A3. Permanently lower GDP growth 1/	96	95	95	95	97	97	103	122
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	96	100	106	108	112	114	125	132
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	96	95	97	96	96	95	90	78
B3. Combination of B1-B2 using one half standard deviation shocks	96	96	99	100	103	105	111	115
B4. One-time 30 percent real depreciation in 2014	96	121	116	111	108	104	89	69
B5. 10 percent of GDP increase in other debt-creating flows in 2014	96	119	117	115	115	113	105	86
Debt Service-to-Revenue Ratio 2/								
Baseline	8	7	7	7	7	7	6	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	7	8	7	7	7	6	6
A2. Primary balance is unchanged from 2013	8	7	7	7	7	7	6	6
A3. Permanently lower GDP growth 1/	8	7	8	7	7	7	6	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	8	7	8	8	8	8	7	8
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	8	7	7	7	7	7	6	5
B3. Combination of B1-B2 using one half standard deviation shocks	8	7	8	8	8	7	7	7
B4. One-time 30 percent real depreciation in 2014	8	8	10	9	9	9	8	7
B5. 10 percent of GDP increase in other debt-creating flows in 2014	8	7	8	8	8	7	6	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

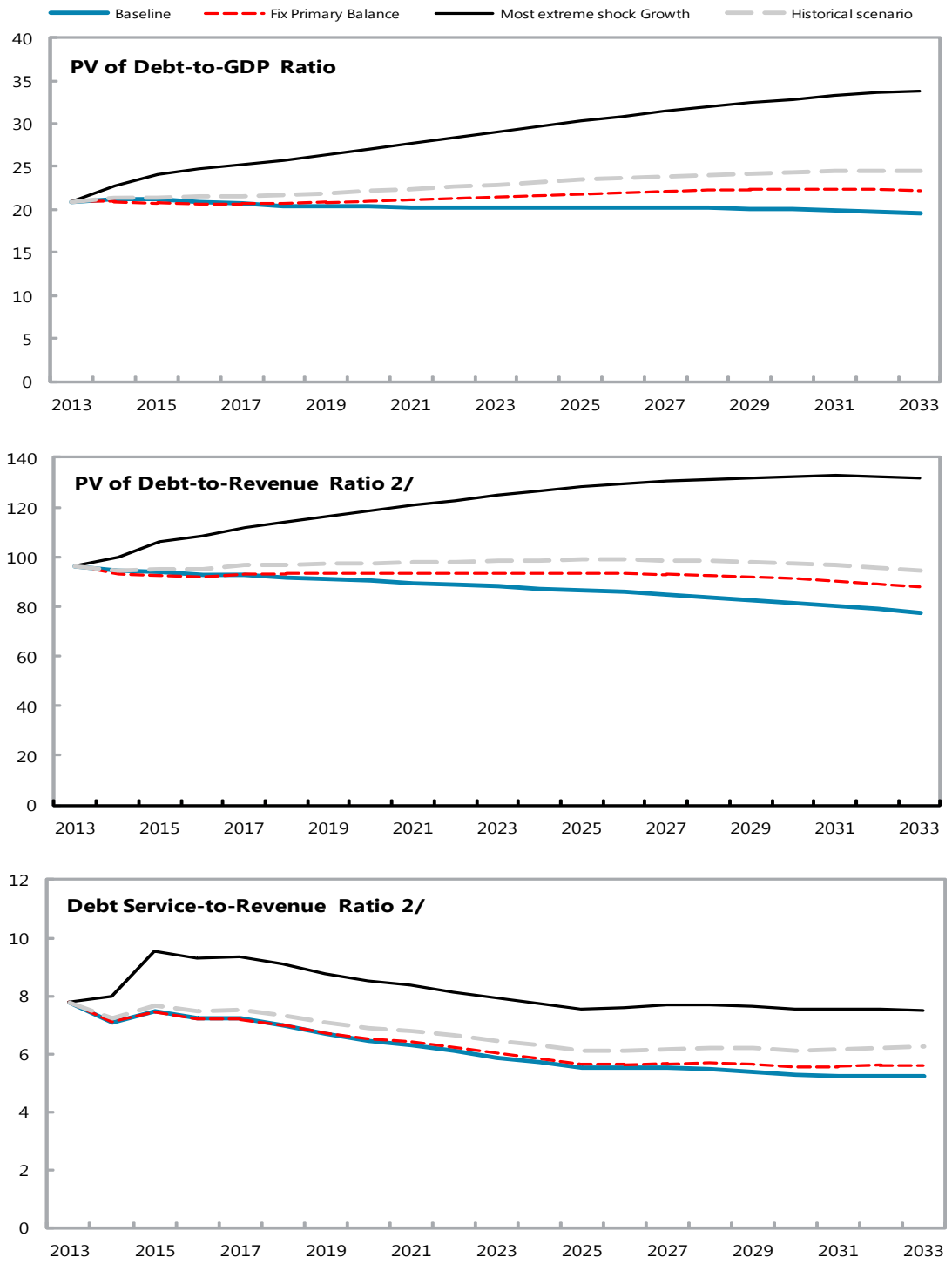
Figure 3. Mali: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2024. In figure b. it corresponds to a Non-debt flows shock; in c. to a Terms shock; in d. to a Non-debt flows shock; in e. to a Terms shock and in figure f. to a One-time depreciation shock

Figure 4.Mali: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 3a.Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033
(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to-GDP+remittances ratio								
Baseline	16	17	17	17	17	17	**17**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	16	15	14	14	13	12	**8**	5
A2. New public sector loans on less favorable terms in 2013-2033 2	16	18	19	20	20	21	**24**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	16	17	18	18	18	18	**19**	19
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	16	18	19	19	19	19	**19**	19
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	16	18	18	18	18	18	**19**	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	16	24	28	27	26	26	**25**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	16	24	27	26	26	25	**25**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	16	23	23	23	23	23	**24**	25
PV of debt-to-exports+remittances ratio								
Baseline	53	59	64	68	72	74	**84**	92
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	53	55	55	55	54	53	**40**	27
A2. New public sector loans on less favorable terms in 2013-2033 2	53	63	71	79	86	92	**118**	153
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	53	60	64	69	72	74	**85**	94
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	53	65	77	83	86	89	**99**	104
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	53	60	64	69	72	74	**85**	94
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	53	94	113	108	112	114	**121**	109
B5. Combination of B1-B4 using one-half standard deviation shocks	53	88	98	95	99	101	**109**	101
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	53	60	64	69	72	74	**85**	94
PV of debt-to-revenue ratio								
Baseline	95	98	95	93	93	92	**92**	84
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	95	91	82	75	69	63	**40**	21
A2. New public sector loans on less favorable terms in 2013-2033 2	95	105	106	108	111	114	**128**	139
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	95	103	103	101	101	100	**100**	93
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	95	103	106	104	103	102	**100**	88
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	95	103	104	102	101	101	**100**	93
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	95	136	153	147	144	141	**132**	99
B5. Combination of B1-B4 using one-half standard deviation shocks	95	135	149	144	141	139	**131**	102
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	95	138	134	132	131	130	**130**	120

Table 3a.Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (Continued)
(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
Debt service-to-exports+remittances ratio								
Baseline	3	3	4	4	4	4	**5**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	3	3	4	4	4	4	**3**	2
A2. New public sector loans on less favorable terms in 2013-2033 2	3	3	4	4	5	5	**6**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	3	3	4	4	4	4	**5**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	3	3	4	5	5	5	**5**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	3	3	4	4	4	4	**5**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	3	4	5	5	5	5	**5**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	5	5	5	5	**5**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	3	3	4	4	4	4	**5**	6
Debt service-to-revenue ratio								
Baseline	5	5	6	6	6	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	5	5	5	5	5	4	**3**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	5	5	6	6	6	7	**7**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	5	6	6	6	6	6	**6**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	5	6	6	6	6	**5**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	5	6	6	6	6	6	**6**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	5	5	7	7	7	7	**6**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	6	7	7	7	7	**6**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	5	7	8	8	8	8	**7**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	**44**	44

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.